

2.1.3 About A Router

A router (the “rou” in “router” rhymes with “cow”) is an electronic device which comes in a little box, and which connects one network with another (for example, a Local Area Network with a Wide Area Network). Routers do most of the task of directing traffic on the Internet by sending the request from your network to another while simultaneously preventing other unauthorised users to access your network (this latter task is done with the help of a firewall, a software program that intercepts and filters unwanted packets of data that are either trying to enter your network or access the resources of your network). A router examines each packet of data that travels across the Internet, figures out where it is headed, and “routes” it towards its destination in the most efficient way possible. Usually, the packet of data goes to another router and another, and another—until it reaches the destination. Each router that a packet of data passes through to reach its destination is considered a “hop.” A router must depend, however, on gateways.

2.1.4 Gateways And Bridges

Just when you—finally—figured out that “windows” aren’t something you open to let fresh air in, these computer guys tell you to start using “gateways,” too. And you got it right this time—a “gateway” is not something you open to let people in.

A gateway is a network point—a node—that acts as an entrance to another network. It is basically a device that allows the exchange of data by translating between the protocols of two different networks or computers on the connected networks. In the case of your residential connection, the gateway is your ISP. Gateways are different from Bridges in the sense that a bridge only links LANs together, sending data to another LAN while keeping local data within its own network.

Along the way, the data we transact—apart from passing through several routers, gateways, and backbones—is also dependent on an indispensable device called a Server. Servers are of different types, and they perform several important functions.